

▼ **Figure 51.7 Imprinting.** Young graylag geese imprinted on a man.



**WHAT IF?** Suppose the geese were bred to each other. How might their imprinting on a human affect their offspring? Explain.

➔ **Mastering Biology Video: Ducklings**

Therefore, an organism's fitness may be enhanced by the capacity for **spatial learning**, the establishment of a memory that reflects the environment's spatial structure.

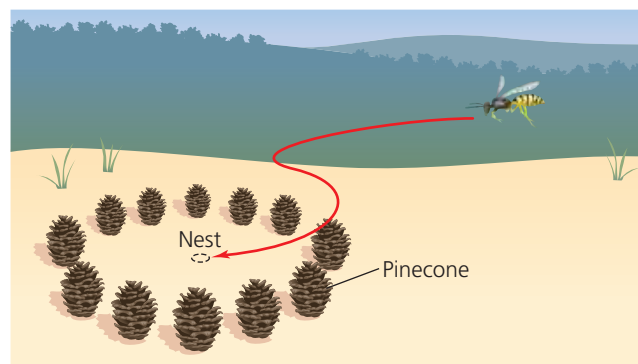
The idea of spatial learning intrigued Tinbergen while he was a graduate student in the Netherlands. At that time, he was studying females of a digger wasp species (*Philanthus triangulum*) that nests in small burrows dug into sand dunes. When a wasp leaves her nest to go hunting, she hides the entrance from potential intruders by covering it with sand. When she returns, however, she flies directly to her hidden nest, despite the presence of hundreds of other burrows in the area. How does she accomplish this feat? Tinbergen hypothesized that a wasp locates her nest by learning its position relative to visible landmarks. To test his hypothesis, he carried out an experiment in the wasps' natural habitat (**Figure 51.8**). By manipulating objects around nest entrances, he demonstrated that digger wasps engage in spatial learning. This experiment was so simple and informative that it could be summarized very concisely. In fact, at 32 pages, Tinbergen's Ph.D. thesis from 1932 is still the shortest ever approved at Leiden University.

In some animals, spatial learning involves formulating a **cognitive map**, a representation in an animal's nervous

▼ **Figure 51.8 Inquiry**

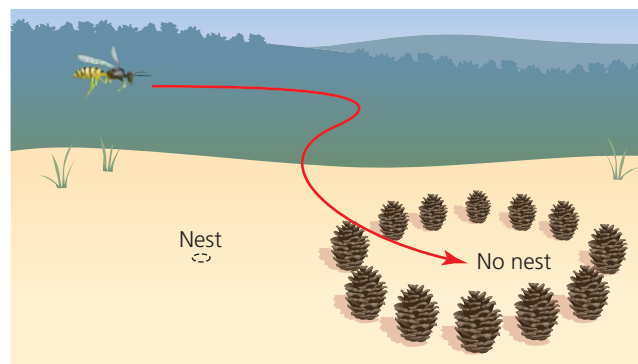
### Does a digger wasp use landmarks to find her nest?

**Experiment** A female digger wasp covers the entrance to her nest while foraging for food, but finds the correct wasp nest reliably upon her return 30 minutes or more later. Niko Tinbergen wanted to test the hypothesis that a wasp learns visual landmarks that mark her nest before she leaves on hunting trips. First, he marked one nest with a ring of pinecones while the wasp was in the burrow. After leaving the nest to forage, the wasp returned to the nest successfully.



Two days later, after the wasp had again left, Tinbergen shifted the ring of pinecones away from the nest. Then he waited to observe the wasp's behavior.

**Results** When the wasp returned, she flew to the center of the pinecone circle instead of to the nearby nest. Repeating the experiment with many wasps, Tinbergen obtained the same results.



**Conclusion** The experiment supported the hypothesis that digger wasps use visual landmarks to keep track of their nests.

**Data from** N. Tinbergen, *The Study of Instinct*, Clarendon Press, Oxford (1951).

**WHAT IF?** Suppose the digger wasp had returned to her original nest site, despite the pinecones having been moved. What alternative hypotheses might you propose regarding how the wasp finds her nest and why the pinecones didn't misdirect the wasp?

➔ **Mastering Biology Animation: Digger Wasps and Landmarks**